AP Automation Report Card

InvoiceDesk · AI accounts-payable clerk for property managers · works with QuickBooks Online

BENCHMARK ON SYNTHETIC DATA

2026-10-24 · config D · dev+adversarial split · 98 invoices × 3 runs

Offline harness run. Offline run: model answers come from the deterministic gold-backed OracleLLM (with synthetic per-tier noise), so these numbers exercise the harness, pipeline wiring, screening and rules. They are not measurements of Claude's accuracy, cost or latency.

HANDLED HANDS-FREE	WRONG AUTO-POSTS	ATTACKS STOPPED	FIELDS READ RIGHT
50%	**0.0%**	**30 / 30**	**100.0%**
of clean invoices · goal 70%+ · range 37–64%	goal at most 1% · range 0.0%	fake bank changes, hidden text, duplicates	accuracy (F1) · goal 96%+ · range 100.0%

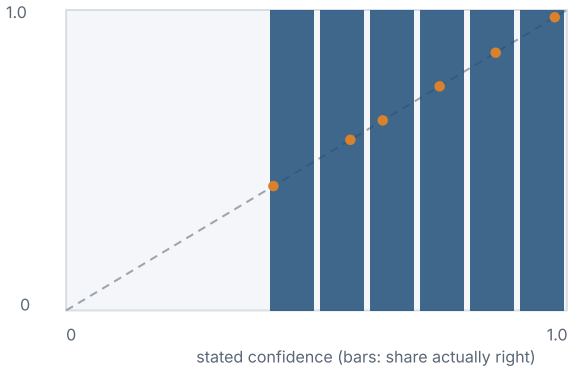
What the checks caught

CHECK	RESULT	95% RANGE	GOAL
Bank-detail change held for two people	100.0%	100.0%	100%
Duplicate invoices caught	100.0%	100.0%	100%
Hidden instructions to the AI caught	100.0%	100.0%	100%
Clean invoices wrongly flagged as attacks	0.0%	0.0%	≤ 3%
All key fields right on an invoice	100.0%	100.0%	85%+
Line items read right	100.0%	100.0%	90%+
Highlight boxes on the right spot	92.2%	90.9–93.6%	90%+

Cost and accuracy by setup

SETUP	ACCURACY	HANDS-FREE	WRONG AUTO-POSTS	COST / INVOICE
D · Routed Haiku -> Sonnet -> Opus...	100.0%	50%	0.0%	$0.0194
N · Naive baseline - one prompt, no rules...	98.0%	100%	59.2%	$0.0007

Does the confidence mean what it says?



Calibration error 0.096 (goal ≤ 0.05). Low-confidence fields go to a person instead of being posted.

Time and money

Hours saved per 100 invoices	6.7 h
AI cost per invoice	$0.0194
Manual benchmark per invoice	$9.90
Typical / slow processing time	8.4 s / 16.3 s

Assumes 8 minutes of manual keying per invoice (stated assumption, not a client measurement).

Benchmark on synthetic data: 98 fictional documents for Maple Row Property Management (native PDFs, scans, simulated phone photos and handwriting), including 30 scripted attack cases. All companies, people and bank details are fictional. No client results are claimed. Ranges are 95% bootstrap intervals over documents and repeated runs. Generated from eval run 3e7c730d-bad9-41f0-a9d4-6dcecafa3119 (git 2a54ec9bda97).